LungLogic: Model-Based Airflow Control with Measured-Delay Compensation and Auto PID

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1. Pneumatic System & Requirements

OBJECTIVE | Reproduce a physiologic breathing pattern in one 250 mL lung branch.

Design targets combine respiratory timing, tidal volume and repeatability despite valve nonlinearity, sensor dynamics and real pneumatic delay.

Respiratory rate	**12–16 bpm**
Tidal volume	**250 mL / lung**
I:E ratio	**1:2**
Repeatability target	**< ±5%**

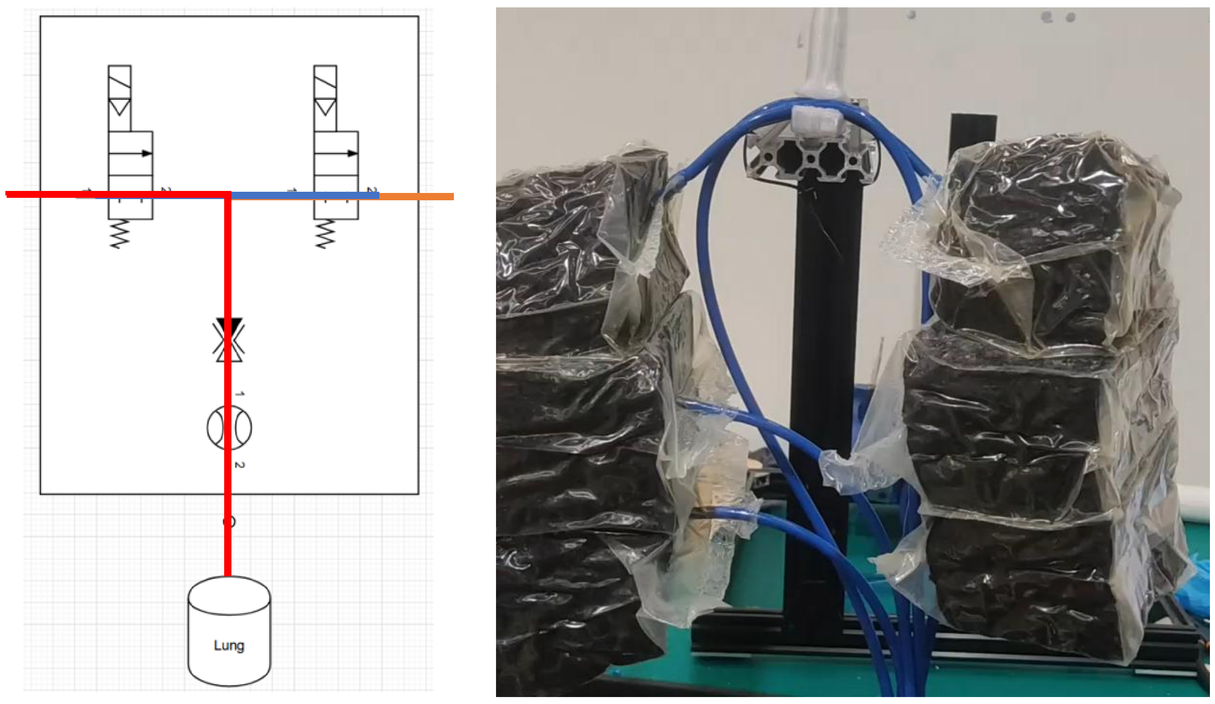


Fig. 1. Single-lung pneumatic route and six-lobe breathing phantom.

2. Embedded Hardware Architecture

OBJECTIVE | Connect software commands to safe, deterministic pneumatic actuation.

The host configures and logs experiments; the embedded controller executes the 100 Hz state machine and flow loop through a protected 24 V bridge.

PC / Python GUI
Setpoints, telemetry, Auto PID

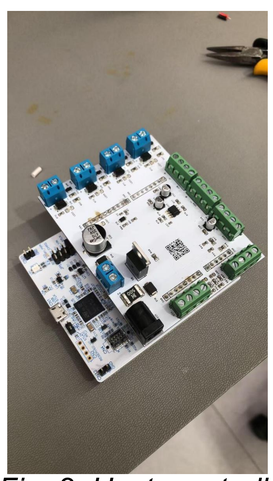
Configuration / telemetry

Embedded controller
State machine and flow feedback

24 V bridge PCB
Power and protected interfaces

Valves
ATM, vacuum and PPV

Sensors
Flow feedback and volume estimate



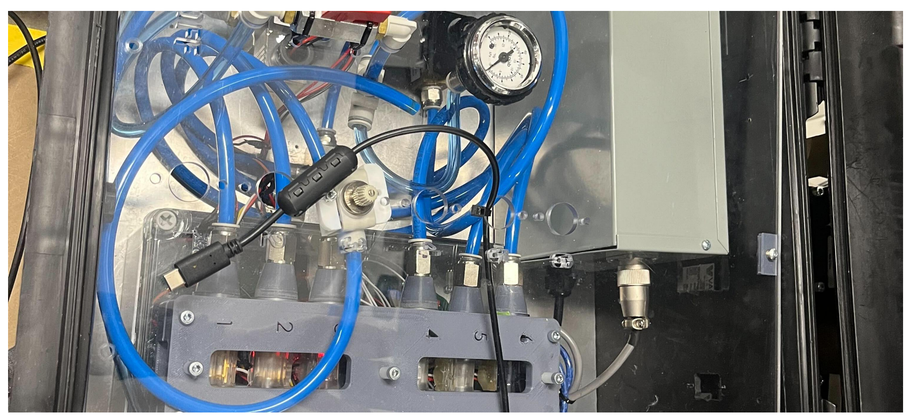


Fig. 2. Host–controller–PCB architecture and pneumatic control case.

3. Control Software & Workflow

OBJECTIVE | Provide a visible, adjustable tool for operation, tuning and traceable experiments.

User Mode simplifies routine runs; Developer Mode exposes PID, model, measured-delay and calibration controls.

User Mode

Developer Mode

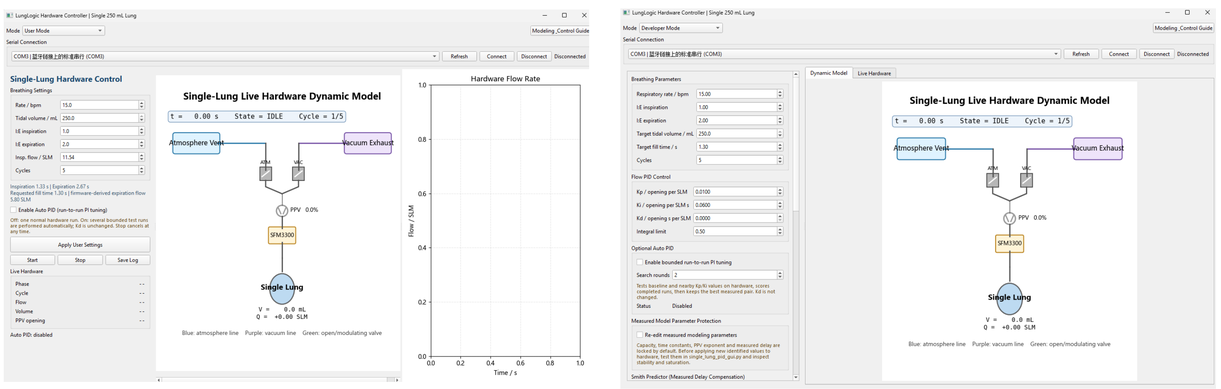


Fig. 3. Telemetry-driven dynamic model and protected developer controls.

Breathing setpoints	**Routine experiment setup**
Dynamic Model	**Live telemetry visualisation**
PID / Smith / Auto PID	**Controller development**
Measured parameters	**Locked after identification**
CSV logging	**Traceable experiment review**

1 Identify the path
Full-open ATM/VAC response
PPV sweep + synchronized delay
Measured model

2 Pre-validate digitally
Test tidal error, tracking, saturation and sensitivity
Reviewed parameter set

3 Deploy and lock
Apply in Developer Mode
short conservative hardware run
Traceable evidence

Workflow: identify → simulate → lock/deploy. Measured model and delay values are protected by default; current scope is one 250 mL branch.

4. Control Method & Validation

OBJECTIVE | Track target flow and tidal volume despite nonlinear PPV response and measured delay.

Feedforward supplies nominal valve opening; measured-flow PID removes residual error; Smith prediction is optional.

Breathing target
RR · VT · I:E
fill time

Reference + routing
Qref(t)
ATM / VAC state machine

Feedforward + PID
aff from identified model
PID corrects flow error

Physical path
PPV · lung · SFM3300
real delay + dynamics

Qfeedback

Optional Smith predictor
identified model + measured delay

Qmeas(t)

$$Q_{ref,in} = 0.06\,V_T/T_{fill} \qquad e = Q_{ref} - |Q_{feedback}| \qquad a = \mathrm{clip}(a_{ff} + K_P e + K_I \int e\,dt + K_D[-d|Q_{feedback}|/dt],\ 0,\ 1)$$

Auto PID: bounded run-to-run PI search

Optional tuning evaluates a stable baseline and bounded Kp/Ki candidates over complete hardware runs. Flow error, tidal error and PPV saturation form the score. Kd and identified model/delay parameters remain fixed; the user reviews the best tested gains.

Measured-delay compensation

Hardware adds no artificial delay. Optional Smith prediction uses the identified first-order model and measured command-to-feedback latency. Re-identify after changing tubing, valves, sensor filtering or sampling.

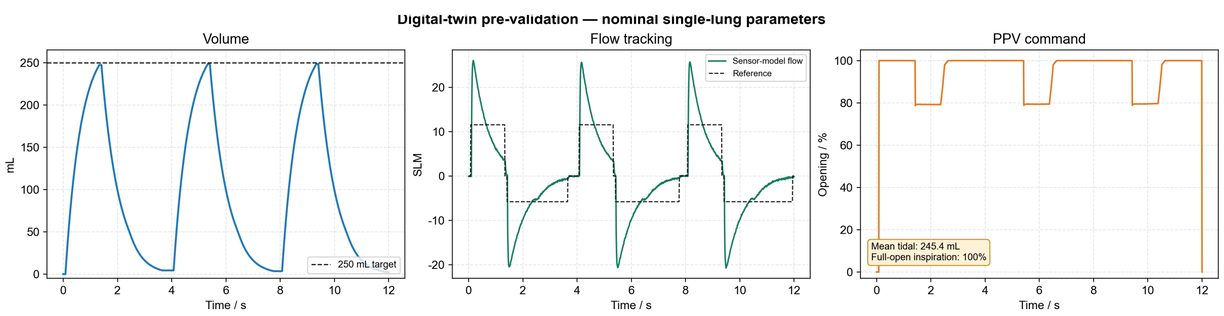


Fig. 4. Digital-twin pre-validation at 15 bpm. Model-based result; measured hardware validation remains required.

Industry collaborator: 4DMedical | Prototype images: project scope deck (19 March 2026) | Simulation result: current single-lung model | Bench prototype—not for clinical use.